

Database Setup

Let's first create a sample database and table for our exercises.

```
CREATE DATABASE company_db;
```

```
USE company_db;
```

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY AUTO_INCREMENT,  
    emp_name VARCHAR(50),  
    department VARCHAR(50),  
    salary DECIMAL(10,2),  
    joining_date DATE  
);
```

```
INSERT INTO employees (emp_name, department, salary, joining_date) VALUES  
( 'Alice', 'HR', 50000, '2021-01-15'),  
( 'Bob', 'IT', 65000, '2020-07-01'),  
( 'Charlie', 'Finance', 55000, '2022-03-12'),  
( 'David', 'IT', 70000, '2021-09-10'),  
( 'Eve', 'Finance', 60000, '2019-11-23'),  
( 'Frank', 'HR', 52000, '2022-02-01'),  
( 'Grace', 'IT', 75000, '2023-06-19'),  
( 'Helen', 'Finance', 58000, '2020-10-30'),  
( 'Ian', 'HR', 49000, '2018-04-25'),  
( 'Jack', 'IT', 72000, '2023-02-14');
```

Exercise 1: Retrieve all employee details

Question: Show all columns from the employees table.

Solution:

```
SELECT * FROM employees;
```

Exercise 2: Display employees from IT department

Question: List names and salaries of all employees working in the IT department.

Solution:

```
SELECT emp_name, salary
FROM employees
WHERE department = 'IT';
```

Exercise 3: Find employees earning more than 60,000

Question: Get all employees whose salary is greater than 60,000.

Solution:

```
SELECT emp_name, department, salary
FROM employees
WHERE salary > 60000;
```

Exercise 4: Sort employees by salary in descending order

Question: Display employee names and salaries sorted from highest to lowest salary.

Solution:

```
SELECT emp_name, salary
FROM employees
ORDER BY salary DESC;
```

Exercise 5: Find total salary paid to each department

Question: Show total salary of each department using GROUP BY.

Solution:

```
SELECT department, SUM(salary) AS total_salary
FROM employees
GROUP BY department;
```

Exercise 6: Find average salary of IT department

Question: What is the average salary of all employees in the IT department?

Solution:

```
SELECT department, AVG(salary) AS avg_salary
FROM employees
WHERE department = 'IT'
GROUP BY department;
```

Exercise 7: Find employees who joined after 2021

Question: Show employee names who joined after 1st Jan 2021.

Solution:

```
SELECT emp_name, joining_date
FROM employees
WHERE joining_date > '2021-01-01';
```

Exercise 8: Display top 3 highest-paid employees

Question: Show top 3 employees with the highest salaries.

Solution:

```
SELECT emp_name, salary
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

Exercise 9: Count number of employees in each department

Question: Find how many employees work in each department.

Solution:

```
SELECT department, COUNT(*) AS total_employees
FROM employees
GROUP BY department;
```

Exercise 10: Find departments with average salary above 55,000

Question: Display only departments where the average salary is greater than 55,000.

Solution:

```
SELECT department, AVG(salary) AS avg_salary
FROM employees
GROUP BY department
HAVING AVG(salary) > 55000;
```